

MSDS 452: Graphical, Network, and Causal Models

Final report

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Causality analysis of student performance factors

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Abstract

I investigated the causal mechanisms underlying student academic performance by applying Causal Bayesian Networks to a dataset of secondary mathematics students from two Portuguese schools. Moving beyond traditional correlation-based predictive modeling, I employed a hybrid strategy that combined automated Causal Structural Discovery (CSD) algorithms with manual refinement guided by human knowledge and iterative d-separation testing. An 11-variable Directed Acyclic Graph (DAG) was constructed and validated against statistical independence constraints. The resulting structural causal model identified critical causal pathways, such as, that parental education directly influences academic failures independent of other factors, and that student motivation (higher education aspiration) drives study time and subsequent performance. The intervention/counterfactual analysis reveals that targeting student motivation would yield an average predicted grade increase of 8.3 points, while increasing study time would improve grades by 4.5 points. Interventional simulations provided actionable insights for means to improve grades, particularly for a student subgroup with prior failures. The results underscore the value of causal inference in transforming observational data into a robust framework for policy simulation and forecasting.

Keywords: Causal Structural Discovery, Structural Causal Model, Directed Acyclic Graph, Intervention Analysis, Student Performance

1.0 Introduction

Understanding the key drivers of student academic performance is the main theme of educational research and policy, with profound implications in shaping effective public spending, providing equal opportunities for the next generation, and ultimately fostering a competitive workforce that powers the economy. A large body of literature has established strong correlations between student success and factors like socio-economic status, past performance, and school environment. While these factors are good predictors of student performance, the underlying statistical models cannot establish causality, let alone draw conclusions on the causal effect of intervention policies. Costly randomized controlled experiments can be performed as a remedy, but they are inefficient and often difficult to set up. This study aims to move beyond correlation-based modeling to investigate the causal mechanisms that determine student achievement using causal inference.

I used a dataset of student math performance measured in grades in secondary education of two Portuguese schools, available from the UCI Machine Learning Repository (Cortez 2008). It contains 649 instances of 30 attributes, capturing a comprehensive profile of students, including their demographic background (e.g., sex, age, parent's education and jobs), social characteristics (e.g., family size, parental cohabitation status), and school-related features (e.g., study time, past failures, extra support). The original study aimed to achieve accurate prediction of student performance with Decision Trees, Random Forest, Neural Networks and Support Vector Machines. Although it identified strong predictors, such as past performance, lifestyle and parent education, this approach struggles to translate predictive insights into actionable interventions. Using causal inference methods, I can identify potential confounders (like family background, measures of motivation) and investigate the causal effect of potential treatment variables, such as extra educational support, intervention on school absence, or promoting student motivation for higher education.

2.0 Literature review

The heart of causal inference is modeling variable relationships with Directed Acyclic Graphs (DAG). The nodes represent variables and the directed edges represent the functions mapping one variable to another. The graph model imposes constraints on the variables so that each variable is independent of its non-descendants given its parents (Markov conditions). The parent (P) to child (C) relationship between nodes expresses the dependency of their probability distribution, $P(P, C) = P(C | P)P(P)$. By applying the chain rule of probability, the joint probability of all variables, $P(data|model)$, can be calculated as the product of the independent Markov kernels. Like all other machine learning algorithms, causal inference aims to find the optimal causal model that best fits the data, that is, maximizes the likelihood or a posteriori. To achieve this, we need to establish the DAG and learn the parameters that determine the conditional dependencies between all P and C.

The initial causal model can be constructed manually based on human intuition and domain knowledge, through causal structure discovery (CSD) algorithms, or a combination of both. The

causal relationships between variables induce independence or conditional independence for d-separations, defined as the path between nodes blocked by colliders (in $A \rightarrow B \leftarrow C$, A and C are separated) or by conditioning on the middle nodes of a chain ($A \rightarrow B \rightarrow C$) or a fork ($A \leftarrow B \rightarrow C$) (Pearl, Glymour and Jewell 2016). By performing independence tests, the current model is either accepted or rejected (Ness 2025). Modifications can be made directly to edges of the failed tests, and further independence tests are carried out for each new model until the irrefutable model is reached.

When human knowledge in an area is limited or biased, and as the dimensionality of datasets grows, CSD provides a viable approach to generate an initial DAG for subsequent testing. The goal of CSD is to find the most optimal Bayesian network with a given set of nodes that represent dependent features. The challenge of this optimization problem is not only that the number of possible network structures grows super-exponentially with the number of nodes, but also that the solution space is discrete. Causal discovery methods are broadly categorized into constraint-based and score-based algorithms, typically applied when the sample size significantly exceeds the number of observed variables (usually $N \gg p$, where $p < 30$). Constraint-based approaches, such as the PC (Peter-Clark) algorithm (Spirtes, Glymour and Scheines 2001) and its latent-variable variants (Spirtes, Meek and Richardson 1995), infer edge presence and direction by strictly utilizing statistical dependence and conditional independence tests. In contrast, score-based algorithms identify the optimal structure by maximizing a Bayesian scoring criterion that measures the local goodness-of-fit for each node relative to its parents. Notable among these is Greedy Equivalence Search (GES) (Chickering 2002), a two-phase process that drastically reduces the search space by navigating Markov equivalence classes rather than individual DAGs. Similar exact search method introduced by Silander and Myllymäki (2006) follows the steps of identifying most likely parents of each node and the best sink node, and reordering the sequences of causal chain to maximize a Bayesian scoring criterion.

CSD algorithms rely on heavy theoretical assumptions, such as causal faithfulness or Meek conjecture, and show limitations in finalizing edge directions, handling latent variable, and sensitivity to data quality and size. Therefore, the strategy for causal modeling should be utilizing causal discovery to narrow down candidate graphs for subsequent independence testing, while blending in human oversight and domain expertise.

The true power of causal models lies in their ability to infer causal effects through intervention and counterfactual analyses. An intervention analysis involves changing a parameter to observe the resulting effect of a particular outcome. Mathematically, this is expressed through the do-operator, $do(X = x)$, which signifies an external manipulation that forces a variable X to take a specific value, regardless of its natural causes. In a DAG, an intervention is visualized as graph surgery, where all incoming edges to the intervened variable are deleted, decoupling it from its ancestors. Pearl (2016) built the mathematical foundation for computing do operations with a parameterized DAG. Practical implementation of intervention analysis has made profound impact on researchers' interpretation of observational data and provide

essential insight in epidemiology, clinical trials and policymaking, where randomized controlled experiments are difficult to carry out.

Counterfactual is a special type of intervention. While interventions are prospective and simulate the future impact of a change on the entire population, counterfactuals are retrospective, aiming to understand an alternative past for a subset of incidences. Instead of using stochastic noise, counterfactuals rely on specific exogenous variables, which represent the actual noise (unpredictable portion of a variable) observed in a real-world event to determine what would have occurred had a set of factors been different. Consequently, counterfactuals provide a more individualized inference by fixing the background conditions of a specific moment in time. The types of counterfactual queries for this dataset are: if a student whose parents have lower education level ($PE = 0$) has the intention to pursue higher education ($HE = 1$), what would their grade be like if their parents had higher education? We can express it as $E[Grade_{PE=} | PE = 0, HE = 1]$. If we intervene on the motivation of students to pursue higher education, what is the average effect on all students (ATE), or what is the expected change in grades for students with intentions for higher education in comparison to the counterfactual that they had no such intentions (ATT, average effect of treatment on the treated), or for students without higher aspiration in comparison to the counterfactual that they had such aspiration (ATU, average effect of treatment on the untreated).

3.0 Methods

3.1 Exploratory Data Analysis and data preparation

I first excluded the age variable, since all cases were in secondary education. The longitudinal grades within one school year (G1, G2, G3) were summed to generate a total `grade` variable.

Two sets of numerical variables that showed strong correlations (Figure 1) were aggregated by summing: `Fedu` (father's education) and `Medu` (mother's education) to generate `Pedu`; `Dalc` (workday alcohol consumption) and `Walc` (weekend alcohol consumption) to generate `alc`. This resulted in 27 predictor variables.

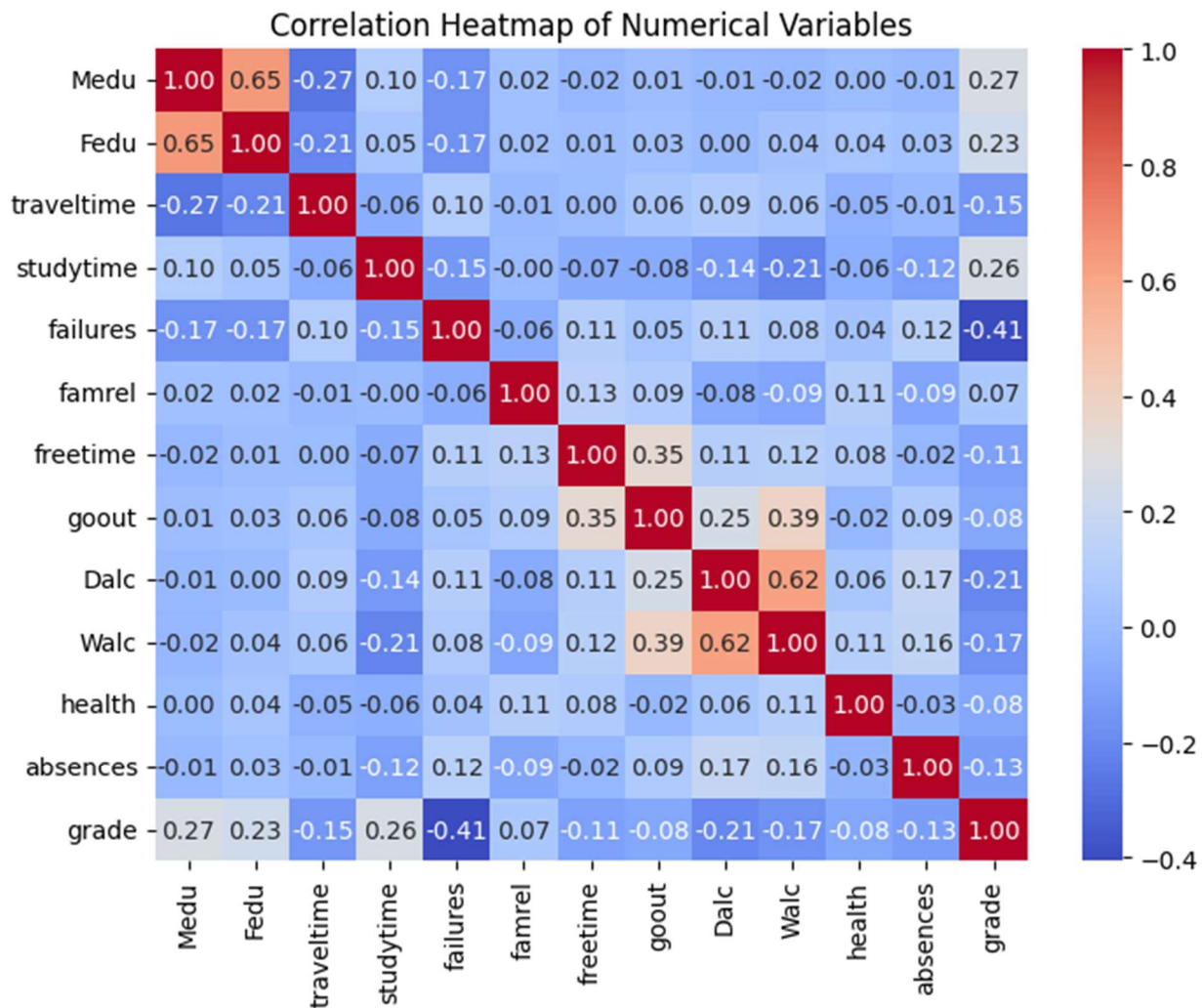


Figure 1 Correlation matrix of numerical features

A random forest regressor was then trained to identify important features. I prioritized features that exhibit a consistent and monotonic relationship with the model's output, as evidenced by clear color separation along the SHAP value axis (Shapley Additive exPlanations, (Lundberg and Lee 2017)). Specifically, I selected variables where high feature values (red) are distinctly clustered on one side of the center line and low values (blue) on the other (Figure 2), indicating a stable contribution to the prediction, with some tolerance of noise for features that are in the top 10 of important features. The selected features are:

```
top_features = ['failures', 'alc', 'Pedu', 'absences', 'higher',
               'studytime', 'school', 'schoolsup', 'address', 'sex']
```

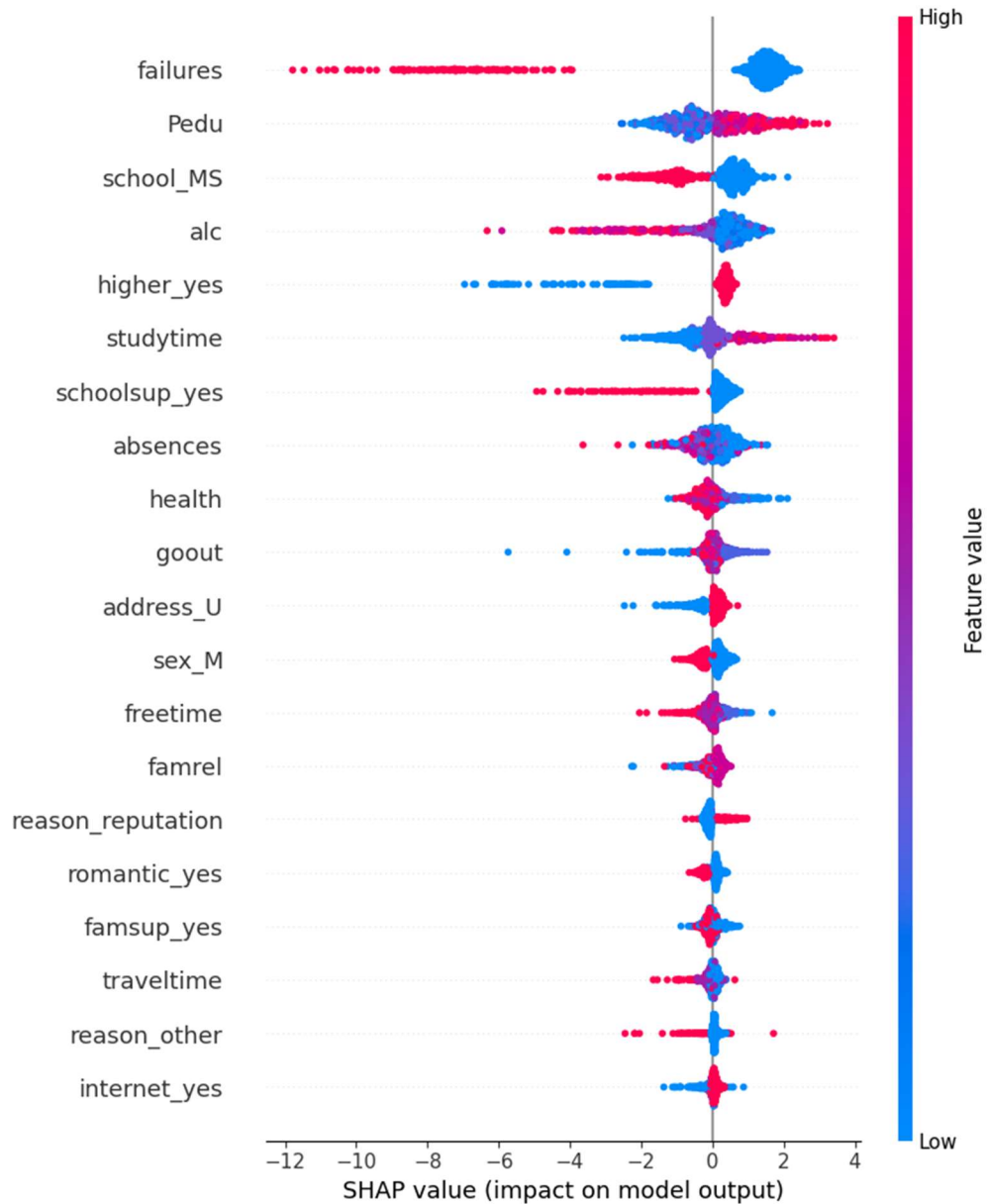


Figure 2 SHAP summary plot of important features

The continuous `grade` variable was discretized to generate `grade_group` with three categories based on its distribution: Low (bottom 20th percentile), High (top 20th percentile), and Mid (the remaining 60%). Variable `absences` was also discretized to generate `absences_group` with two categories: normal (0-5), high (>5). Standardizing the data types across all variables ensures compatibility with specific conditional independence tests for constraint-based algorithms and scoring functions for score-based methods.

3.2 Causal discovery algorithms

I used the Peter-Clark (PC) Stable algorithm and Greedy Equivalent Search (GES) algorithm implemented in `causal-learn` python module (Zheng, et al. 2024). The independent test for

PC was Chi-squared test. The scoring criterion in GES for discrete data is Bayesian Dirichlet equivalent uniform (BDeu) for Dirichlet distribution with uniform prior on graph structure.

3.3 d-separations and independence test

The d-separations were calculated using `get_independencies()` function from the `pgmpy` module v1.0.0 (Ankan and Textor 2024).

The appropriate independence test for categorical variables is chi-squared test. A p-value threshold of 0.05 was chosen to determine dependences between variables: $p < 0.05$, dependence exists and the model is refuted. No family-wise error correction was used.

3.4 Construct structural causal model

A Structural Causal Model (SCM) was initialized using the `gcm.InvertibleStructuralCausalModel` class, with causal mechanisms automatically assigned to each node. The functional relationships for non-root nodes were estimated by systematically evaluating various `scikit-learn` regressors/classifiers, selecting the optimal predictive model based on the node's direct causal parents. The performance of each functional mechanism were evaluated using continuous ranked probability score (CRPS), which measures the absolute difference between the forecasted cumulative distribution function (CDF) and the empirical CDF.

3.5 Intervention and Counterfactual analysis

Average treatment effects (ATE and ATT) were estimated using the standard `dowhy` pipeline: first initializing a `CausalModel` with specified treatment and outcome variables, then determining the estimand via `identify_estimand`, and finally executing the estimation through `estimate_effect`. Specific target populations were analyzed by adjusting the `target_units` parameter. Refutation of the causal effect was also carried out through `refute_estimate` function by either adding a random common cause to the graph, or subsetting the data, or simulate placebo treatment through `do` operations.

Additionally, the `gcm.interventional_samples` utility was employed to simulate post-intervention data distributions. Counterfactual simulation was omitted as the current `gcm` framework does not support exogenous noise inversion for classifier-based functional causal models.

4.0 Results

4.1 Constructing the DAG

The initial PC algorithm output (Figure 3) failed to incorporate `absences_group`, despite significant dependencies identified via pairwise chi-squared tests (Figure 4). Furthermore, the algorithm produced undirected edges between `higher`, `studytime`, and `grade_group`. Given that `higher` (student motivation) is a foundational driver of academic behavior, it should

have been positioned closer to the root of the structure. To address these inferential errors, I introduced the following structural priors to the learning process:

- `higher` $\rightarrow$ `studytime` & `higher` $\rightarrow$ `grade_group`: Directionality reflects that motivation dictates academic effort and outcomes.
- `address` $\rightarrow$ `school`: A constraint based on geographical and administrative logic. Cross tabulation of the two variables showed some level of dependence but not redundancy.
- Edge Exclusion (`grade` $\nleftrightarrow$ `failures`): I enforced the absence of a direct link, assuming their correlation is mediated by latent or underlying factors rather than a direct causal path.
- `grade_group` $\rightarrow$ `schoolsup`: the fact that students with `schoolsup` have worse grades (Figure 2) is counterintuitive. What might be a probable scenario is that students with worse grades were more likely to seek out school support.

The outcome graph (Figure 3, bottom) corrected the above errors but was still inconsistent with the pair-wise independence tests (Figure 4).

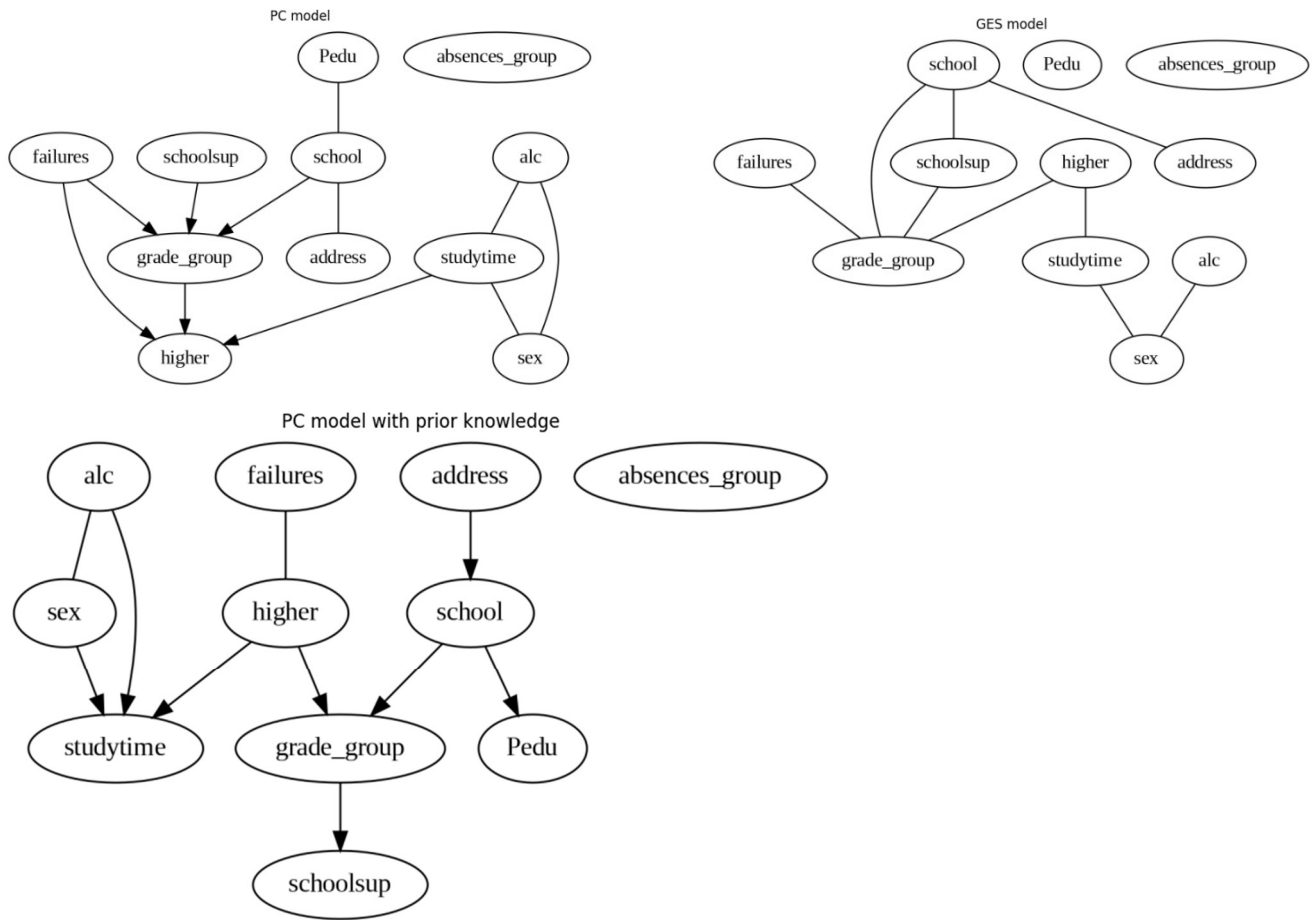


Figure 3 Top left, output DAG of PC. Top right, output DAG of GES. Bottom, output of constrained PC

The GES algorithm produced a Completed Partially Directed Acyclic Graph (CPDAG), as it searches across Markov equivalence classes rather than individual DAGs. While the algorithm was unable to orient all edge directions from the data alone (Figure 3, top right), it confirmed the structural dependencies within two key clusters: (alc, studytime, sex) and (higher, studytime, grade_group). These identified triads served as a helpful starting point for manual structural refinement.

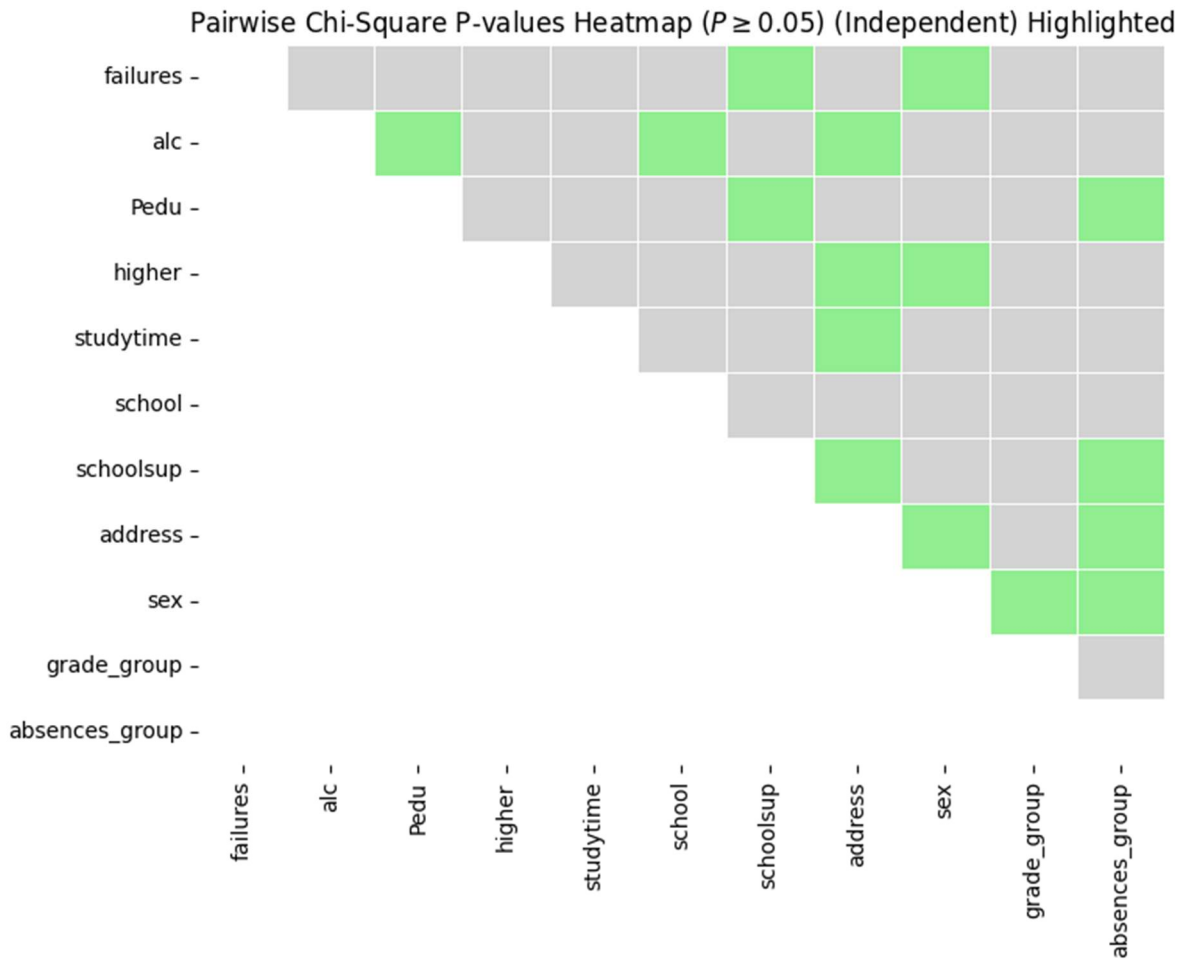


Figure 4 Pair-wise chi-squared test. Independences ($p \geq 0.05$) were highlighted in green.

Starting with the skeleton provided by the constrained PC algorithm (Figure 3) and initial independence tests (Figure 4), I manually constructed a candidate Directed Acyclic Graph (DAG) (Figure 5, left). Using the `pgmpy` library, I iteratively performed independence tests of d-separations to identify discrepancies between the model and the data. After refining the structure to resolve these falsified dependencies, I arrived at a final model (Figure 5, right) that is fully consistent with the observed independence constraints. It reveals some unexpected but interesting causal relationships such as, $\text{Pedu} \rightarrow \text{sex}$, $\text{sex} \rightarrow \text{school}$, and Pedu has a direct link on `failure` that is not through other factors, `school` has a separate path to grades besides the path $\text{school} \rightarrow \text{studytime} \rightarrow \text{grade}$.

The dependency between variable `school` and `failure` did not distinguish the direction of causality (both passed the chi-squared tests for d-separations using the `pgmpy` module). Evaluation of their equivalence through permutation tests using `dowhy.gcm.falsify_graph` function yielded similar TPa p-value (measures the probability of edge-permuted DAG being in the same Markov equivalent class as the given DAG by testing parental d-separations) and LCM p-value (measures the probability of random node permutations that violates Local Causal Mechanism less than the given DAG) for the two

graphs, suggesting that the data supports both causal structures almost equally, with both considered significantly informative than the permutations. Although `falsify_graph` suggested several similar edge removals for both graphs on the ground of causal minimality, none of them passed the independence test (data not shown).

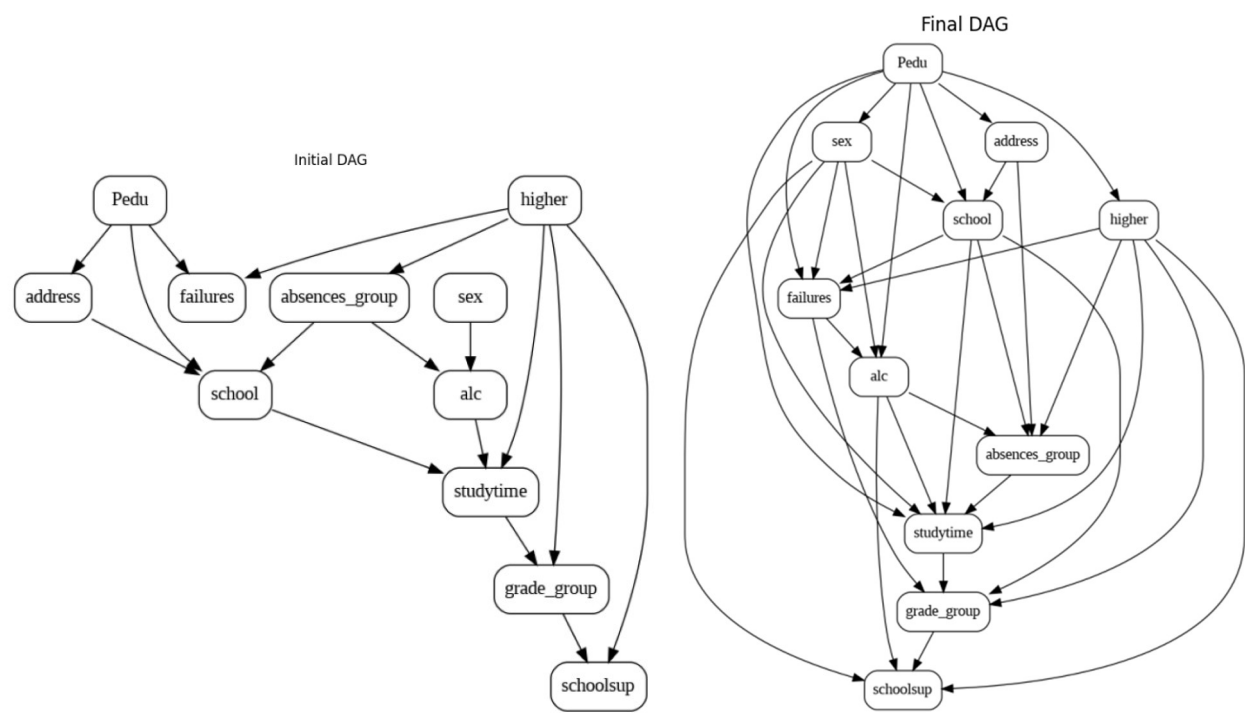
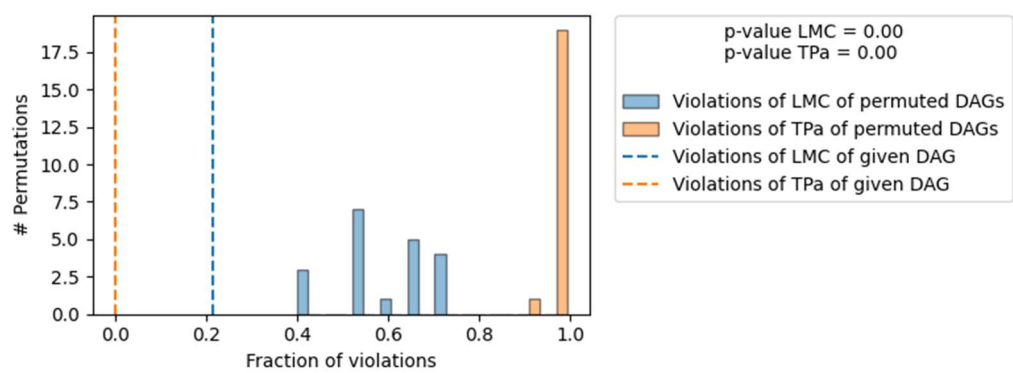


Figure 5 Left, initial manually constructed DAG. Right, final model that is consistent with the data.



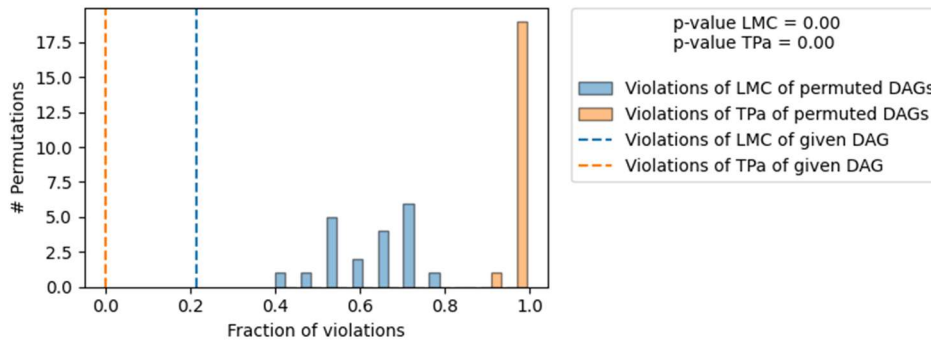


Figure 6 Top, permutation tests of graph with $\text{school} \rightarrow \text{failures}$. Bottom, permutation tests of graph with $\text{failures} \rightarrow \text{school}$.

4.2 Training and evaluation of the structural causal model

The causal mechanisms were defined by its data types: the root node (Pedu) follows an empirical stochastic distribution, while non-root categorical nodes utilize `ClassifierBasedConditionalStochasticModel` class. For ordinal variables like `studytime` and `grade`, `DiscreteAdditiveNoiseModel` structure was applied and fitted with regressor-based functional mechanisms. The CRPS scoring rule indicated good prediction for all categorical non-root nodes, while the discrete nodes were only fair models, with 0.54 CRPS for `studytime` and 0.47 CRPS for `grade`.

The prediction of grades using the submodel of only its direct causal parents had worse performance ($\text{MSE} = 50.5$) than a Gradient Boosted Regressor ($\text{MSE} = 39.4$) using all 10 variables in the causal model as predictors. However, the data generated from the structural causal model (GCM-generated) using `gcm.draw_samples` function had better agreement with the original data distribution than the GBR-predicted values (Figure 7).

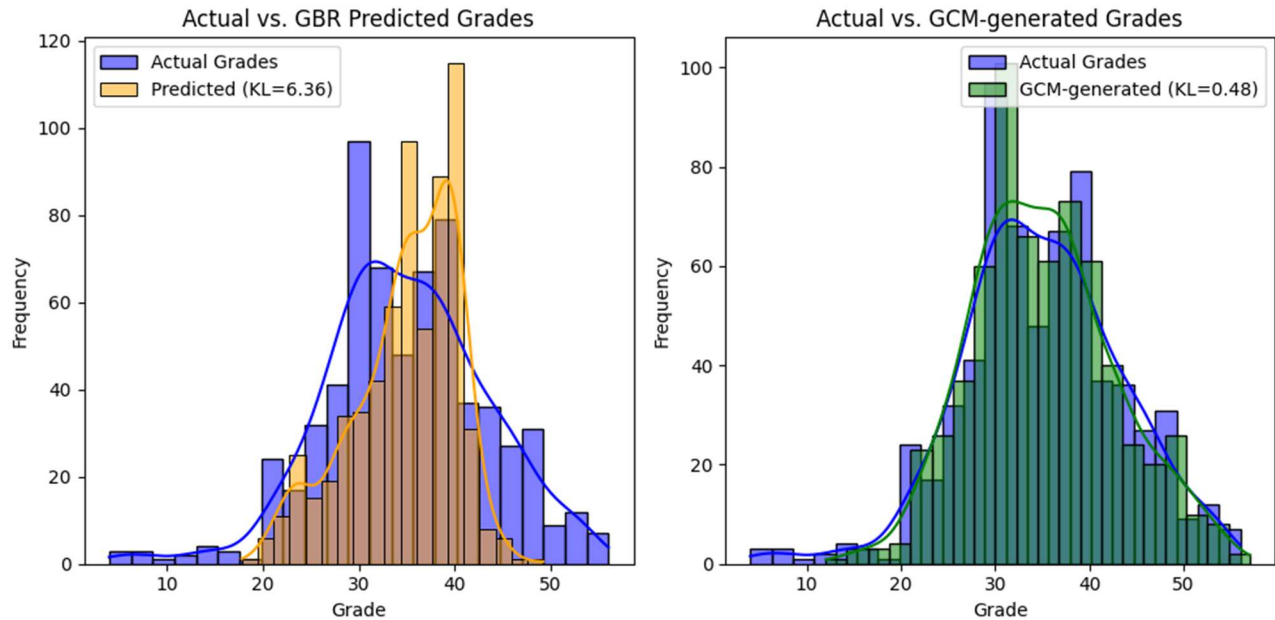


Figure 7 Left, distribution of GBR-predicted grades. Right, distribution of grades generated from the structural causal model.

4.3 Intervention and counterfactual analysis

4.3.1 Causal effect estimation and refutation

I mainly focused on estimating the causal effect of `higher` and `studytime` on `grade`, as summarized in Table 1. The ATE and ATT of `higher` on `grade` are similar, suggesting no treatment selection bias in the samples. If a functional causal model is linear regression as for the relationship between `studytime` and `grade`, the causal effect of the treatment is simply the coefficient of the treatment in the linear regression model. Given their robustness in multiple refute tests, both treatments would have significant impact on students' grades if implemented.

Table 1 ATE and ATT calculations of `higher` and `studytime` on `grade`.

	ATE		ATT	Refute
higher(X) → grade(Y)	$E[Y do(X = 1)] - E[Y do(X = 0)]$		$E[Y_1 - Y_0 X = 1]$	Add a random common cause: Estimated effect:8.34 New effect:8.34 p value:1.0
	Propensity score stratification: Mean value: 8.34 95% CI: [6.16,10.03]	Distance Matching (5-NN): Mean value:10.43 95% CI: [7.35, 14.92]	Propensity score stratification: Mean value: 8.39 95% CI: [6.58,10.19]	Use a subset of data: Estimated effect:8.34 New effect:8.224 p value:0.8
				Use a Placebo Treatment: Estimated effect:8.34 New effect:0.04 p value:1.0
studytime(X) → grade(Y)	$E[Y do(X = 4)] - E[Y do(X = 1)]$		$E[Y_4 - Y_1 X = 4] = \tau$	Add a random common cause: Estimated effect:4.50 New effect:4.50 p value:0.96
	Linear regression: Mean value: 4.5 95% CI: [2.47,6.53]		Same as ATE	Use a subset of data: Estimated effect:4.50 New effect:4.56 p value:1.0
				Use a Placebo Treatment: Estimated effect:4.50 New effect:0.07 p value:0.9

4.3.2 Interventional simulation

An SCM is generative and allows for interventional simulations by explicitly defining the magnitude and nature of change applied to target variables in place of their functional mechanism and propagating that change through the graph. Suppose we are specifically looking for ways to improve grades for students with past failures. We can simulate this intervention for this student group by generating postintervention samples using `gcm.interventional_samples` function. Figure 8 shows simulated outcomes of `grade` after intervening on `higher` ($do(higher = 1)$), `studytime` ($do(ST = ST + 2)$), and combined intervention of `higher` and `studytime`. All shifted their average grade by a significant amount.

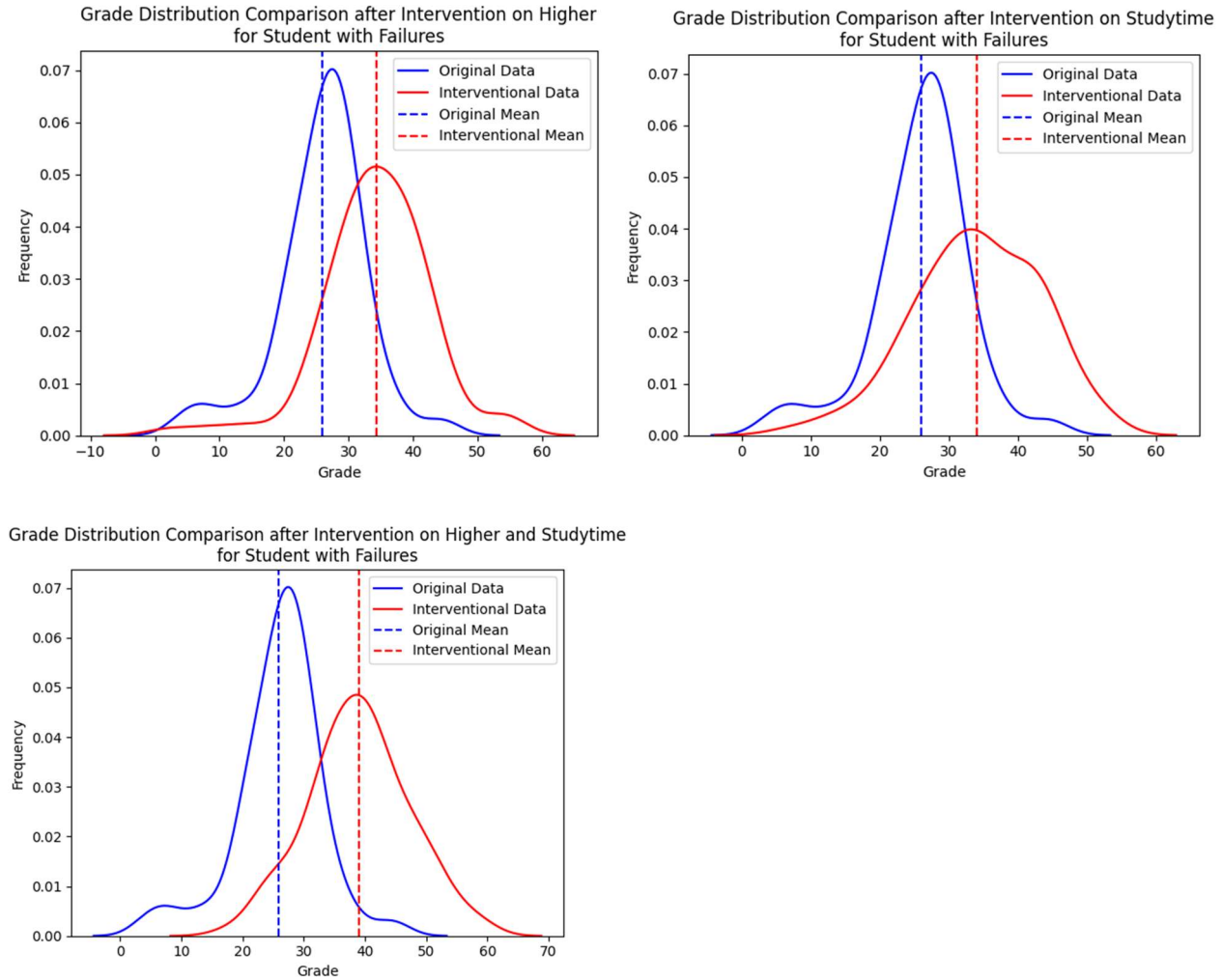


Figure 7 Interventional simulation of grades for students with failures.

5.0 Conclusion

5.1 Structural learning

The groundwork of causal inference starts with constructing the DAG to represent the causal relationships between variables. In this study, I combined CSD and human knowledge to set

up the backbone of the causal mechanism. The causal discovery algorithms did not successfully learn the perfect graph, although it provided a good initial guess of some dependencies in the data. With this 11-variable system, human-guided modifications can achieve consistency with the data. Interestingly, the final graph appears denser than the outputs of causal discovery algorithms, suggesting underlying implicit sparsity restrictions (such as causal minimality) in the tested CSD algorithms.

A failure to reject the null hypotheses during independence testing does not validate the model's correctness; it merely indicates that the current data provides no statistical grounds to dismiss it. This is indicated by the fact that no statistical tests can determine the causal direction between `school` and `failures`. We can use human intuition to determine the likely causal direction is from `school` to `failures`.

Furthermore, the lack of refutation assumes causal sufficiency, the premise that no unobserved common causes are influencing the variables, meaning the discovered graph could still be refuted if hidden confounders exist. During parametric training, the underlying additive noise assumption for `studytime` was rejected through independence test of the noise with its parents, which could mean there is an unobserved confounder for `studytime` and its parent(s).

5.2 Parametric learning

The parametric learning of the SCM in this study was achieved through traditional machine learning models by learning the mapping functions separately between each node and its immediate parents, based on the LMC (Local Markov Condition) assumption. Each model is susceptible to the pitfalls of a regular correlation-based model, such as overfitting to noise in the training data, poor generalization to out-of-distribution data (`grade` submodel prediction range is biased to the middle range), and prediction bias for imbalanced categorical variables (such as for `higher` and `schoolsup`). Misrepresentation of the noise as additive noise can also lead to poor performance (the suboptimal functional mechanism for `studytime` could also mean a more complex noise pattern for this variable). One weak learner among the earlier nodes of the causal hierarchy would propagate the error to all its descendants during generation phase. Neural networks may be a good candidate to improve this for three reasons: they are universal function approximators; the end-to-end differentiability of their architectures allows for joint optimization across the entire causal graph; there is more flexibility to model noise. A viable next step for this project would be embedding causal knowledge into neural network architecture to generate a better SCM (Zhang, et al. 2023).

5.3 Estimate causal effects

Traditional correlation-based modeling identified past failures and parental education as the primary predictors of academic performance. However, these factors are immutable and cannot be candidates for intervention. Causal modeling allows do operations to be evaluated on any estimable variable while bypassing the need for costly, time-consuming randomized controlled trials. Targeting student motivation yields an average predicted increase of 8.3 points for both

the population (ATE) and the treated (ATT), while increasing study time is estimated to improve grades by 4.5 points. These predictions can also be applied specifically to a student subgroup such as students with past failures through intervention simulations. The generative nature of SCMs again strands out over traditional modeling in that it allows simulation of any means of intervention and even their combinations to predict the outcome. From Figure 8, we can propose a combined approach that increases students' motivation and their study time to help students with past failures to get back on track.

In addition, without inherent causal direction in correlation-based modeling, the model considers school support negatively correlated with grades. However, we cannot directly translate this association as school support has negative impact on grades, neither does it support intervention decisions such as reducing school support to increase grades. Causal modeling puts `schoolsup` downstream of `grade` (where low grades trigger support), accounting for the observed negative correlation.

Ultimately, causal modeling transcends simple pattern recognition by providing a structural map of causal relationships between variables, which produces actionable interventional insights and prevents the costly errors from treating correlation as causation.

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